

Wavelet Multi-Index Viewpoint and Experimental Plan

1 Theory of Multi-Index Learning in Wavelet Space

Notation Let $W \in \mathbb{R}^{d \times d}$ be an orthonormal discrete wavelet transform and $c = Wx$ the wavelet coefficients of $x \in \mathbb{R}^d$. Let $\mathcal{J} = \{0, \dots, J\}$ index scales and $\mathcal{B}_j$ the locations in scale j . Write P_j for the orthogonal projector onto subband j in coefficient space. For a kernel matrix $\Theta \in \mathbb{R}^{n \times n}$ define the scale energy

$$\lambda_j := \|P_j \Theta P_j\|_{\text{op}}, \quad \lambda_j^- := \lambda_{\min}(P_j \Theta P_j), \quad \lambda_j^+ := \lambda_{\max}(P_j \Theta P_j).$$

[Orthonormal wavelet transform] W is orthonormal and fixed; inputs are preprocessed as $c = Wx$.

[Wavelet multi-index target] There exist $S \in \mathbb{N}$, vectors $u_s \in \mathbb{R}^d$ each supported on at most s wavelet coordinates, and Lipschitz $g_s: \mathbb{R} \rightarrow \mathbb{R}$ with $\text{Lip}(g_s) \leq L$, such that

$$f^*(x) = \sum_{s=1}^S g_s(\langle u_s, Wx \rangle), \quad \|u_s\|_0 \leq s, \quad \text{supp}(u_s) \subset \bigcup_{j \in \mathcal{J}} \{(j, b) : b \in \mathcal{B}_j\}.$$

We write $s_{\text{tot}} := \sum_{s=1}^S \|u_s\|_0$.

[Wavelet coefficient model] $c = Wx$ has subgaussian coordinates with per-scale normalization $\mathbb{E} c_{j,b}^2 = 1$, and cross-scale covariance is weak: for $j \neq j'$, $\|P_j \Sigma P_{j'}\|_{\text{op}} \leq \varepsilon$ with $\Sigma := \mathbb{E}[cc^\top]$ and small $\varepsilon \geq 0$.

[Lazy training and NTK] Training is in the lazy regime with empirical NTK Θ (two-layer low-rank random features, rank r). Step size ensures linearization validity over the optimization horizon.

[RKHS/NTK invariance under orthonormal transforms] For architectures whose initialization is isotropic in input space and whose NTK depends on inner products, $k(x, y) = k(\langle x, y \rangle, \|x\|, \|y\|)$, we have $k(Wx, Wy) = k(x, y)$ for any orthonormal W . Consequently, the induced RKHS and NTK spectrum are preserved under the change of variables $x \mapsto Wx$.

Proof sketch. If the weight distribution is rotationally invariant, then for any orthonormal W , $\langle w, Wx \rangle \stackrel{d}{=} \langle W^\top w, x \rangle$, preserving the feature distribution and thus the kernel k . The NTK, being a functional of the feature distribution and activation, is unchanged by orthonormal input transforms. $\square$

1.1 Staircase dynamics via scale-wise NTK energies

[Weak cross-band coupling] For the empirical NTK Θ , we have $\|P_j \Theta P_{j'}\|_{\text{op}} \leq \varepsilon_\Theta$ for $j \neq j'$, with ε_Θ sufficiently small relative to $\min_j \lambda_j^-$.

[Timescale separation and staircase learning] Let $y \in \mathbb{R}^n$ be the training labels and f_t the gradient-flow predictor in the NTK regime with kernel matrix Θ at initialization. Decompose the initial residual $r_0 := y - f_0$ into scale components $r_{0,j} := P_j r_0$. Under Assumptions ?? and ??, for all $t \geq 0$,

$$\|P_j(y - f_t)\|_2 \leq \|r_{0,j}\|_2 e^{-\eta \lambda_j^- t} + O\left(\frac{\varepsilon_\Theta}{\lambda_j^-}\right),$$

and the global training error satisfies

$$\|y - f_t\|_2^2 = \sum_{j \in \mathcal{J}} \|r_{0,j}\|_2^2 e^{-2\eta \lambda_j^- t} + O\left(\varepsilon_\Theta \sum_j \frac{\|r_{0,j}\|_2^2}{\lambda_j^-}\right).$$

If the scale energies are ordered $\lambda_0^- \geq \lambda_1^- \geq \dots \geq \lambda_J^-$, then the loss exhibits staircase drops at times $t_j \asymp 1/(\eta \lambda_j^-)$ from coarse to fine scales.

Proof sketch. In the NTK linearization, $y - f_t = e^{-\eta t \Theta}(y - f_0)$. Approximate Θ by its block-diagonal part $\oplus_j P_j \Theta P_j$ and control off-diagonal leakage by ε_Θ . Each block contracts independently at rate λ_j^- up to an $O(\varepsilon_\Theta / \lambda_j^-)$ error. $\square$

1.2 Generalization with wavelet sparsity

Define the wavelet multi-index RKHS norm proxy $B := L \sum_{s=1}^S \|u_s\|_2$. Consider kernel ridge regression with ridge $\gamma \geq 0$ in the NTK.

[Sample complexity for wavelet multi-index targets] Under Assumptions ??-??, suppose labels are $y_i = f^*(x_i) + \xi_i$ with ξ_i subgaussian noise of variance σ^2 . Let $\hat{f}$ be the kernel regressor with the empirical NTK and ridge $\gamma \in [0, \bar{\gamma}]$. Then with probability at least $1 - \delta$ over the sample and RF randomness,

$$\mathbb{E}[(\hat{f}(x) - f^*(x))^2] \leq C_1 \frac{B^2 S s_{\text{tot}} \log(ed/s_{\text{tot}}) + \sigma^2 \log(1/\delta)}{n} + C_2 \underbrace{\|\Theta - \mathbb{E}\Theta\|_{\text{op}}}_{\text{finite-width deviation}} + \text{AppErr},$$

where AppErr is the approximation error of f^* in the NTK RKHS (zero if f^* lies in the closure). Constants depend on $\bar{\gamma}$ and subgaussian parameters.

Proof sketch. Combine RKHS generalization bounds via effective dimension or localized Rademacher complexity with a sparsity-aware covering of the class $\{x \mapsto g(\langle u, Wx \rangle)\}$ using Lipschitz g and s -sparse u , yielding the $S s_{\text{tot}} \log(ed/s_{\text{tot}})/n$ factor. Add a stability/perturbation term for replacing the population kernel by the empirical NTK. $\square$

1.3 Concentration and spiked-band outlier

[NTK concentration in wavelet space] In the doubly-random Gram regime of the low-rank RF model with rank r , for fixed inputs we have entrywise variance $\text{Var}[\Theta_{ij}] = O(r^{-2})$ and

$$\|\Theta - \mathbb{E}\Theta\|_{\text{op}} = O_{\mathbb{P}}\left(\frac{\sqrt{n} + \sqrt{\log n}}{r}\right).$$

These rates are invariant to orthonormal W .

Proof sketch. Each NTK entry is an average over r rank-1 contributions with an additional $1/r$ normalization from the low-rank bottleneck, yielding $O(r^{-2})$ variance. A matrix Bernstein bound gives the operator-norm rate. Orthonormal W leaves moments unchanged. $\square$

[Spiked wavelet band] There exists a scale j^* and direction v^* supported on subband j^* such that the population kernel restricted to j^* has a spike of strength $\beta > 0$: $P_{j^*}\mathbb{E}\Theta P_{j^*}$ has top eigenvalue $\lambda_{\text{bulk}} + \beta$ separated from the bulk edge λ_{bulk} .

[Single outlier aligned with the active band] Under Assumptions ??, ??, and ??, if β exceeds the bulk edge by a constant margin and n is large enough for the Marchenko–Pastur regime, then with high probability the empirical NTK spectrum exhibits a single outlier $\lambda_{\text{max}}(\Theta)$ separated from the bulk, and its eigenvector correlates with v^* by a constant bounded away from zero. Consequently, the earliest staircase drop coincides with the activation of the spiked band j^* .

Proof sketch. Treat Θ as a spiked deformation of a block-MP bulk. Standard spiked random matrix results imply outlier emergence and eigenvector alignment when the spike exceeds a threshold. Weak cross-band coupling preserves the identification with the spiked subband. $\square$

1.4 Partial training and NTK magnitude

[Factor-of-2 NTK under partial training] For the two-layer model with isotropic initialization and separable parameter blocks (input weights vs. output weights), the full NTK decomposes as $\Theta = \Theta_{\text{in}} + \Theta_{\text{out}}$ with $\mathbb{E}\Theta_{\text{in}} = \mathbb{E}\Theta_{\text{out}}$. Training only the output layer yields NTK $\Theta_{\text{partial}} = \Theta_{\text{out}}$, hence $\mathbb{E}\Theta_{\text{partial}} = \frac{1}{2}\mathbb{E}\Theta$ and all kernel eigenvalues are halved in expectation.

Proof sketch. Under isotropy and independence, input- and output-parameter contributions to the NTK are equal in distribution and uncorrelated at initialization; freezing one block removes its contribution. $\square$

[Practical implications] Theorems ??–?? predict (i) coarse-to-fine staircase with times $t_j \approx (\eta \lambda_j^-)^{-1}$, (ii) reliable kernel regression at moderate rank via $O(r^{-2})$ concentration (Theorem ??), and (iii) a persistent single outlier aligned with the currently active wavelet band. Proposition ?? explains doubled effective learning rate under partial training.